

Arjuna JEE 2025

Maths

DPP: 5

Trigonometric Identities

Q1 Proof

$$\frac{\sin 7A - \sin A}{\sin 8A - \sin 2A} = \cos 4A \sec 5A.$$

Q2 Proof

$$\frac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A} = \cot(A + B) \cot(A - B)$$

Q3 Prove that

$$\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \frac{\tan(A+B)}{\tan(A-B)}.$$

Q4 Proof :

$$\frac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A} = \tan(A - B).$$

Q5 Proof :

$$\cos(A + B) + \sin(A - B) = 2 \sin + B).$$

$$(45^\circ + A) \cos(45^\circ$$

Q6 Proof :

$$\frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \frac{\sin A}{\cos 2A \cos 3A}.$$

Q7 Proof :

$$\frac{\sin(4A - 2B) + \sin(4B - 2A)}{\cos(4A - 2B) + \cos(4B - 2A)} = \tan(A + B)$$

Q8 Proof :

$$\frac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = 4 \cos 2\theta \cos 4\theta.$$

Q9 Proof :

$$\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A.$$

Q10 Proof :

$$\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0.$$

Q11 Proof :

$$\sin 10^\circ + \sin 20^\circ + \sin 40^\circ + \sin 50^\circ = \sin 70^\circ + \sin 80^\circ.$$

Q12 Proof :

$$(\sin 3A + \sin A) \sin A$$

$$+ (\cos 3A - \cos A) \cos A = 0$$

Q13 Proof :

$$\frac{2 \sin(A - C) \cos C - \sin(A - 2C)}{2 \sin(B - C) \cos C - \sin(B - 2C)} = \frac{\sin A}{\sin B}.$$

Q14 Proof :

$$\frac{\sin A \sin 2A + \sin 3A \sin 6A + \sin 4A \sin 13A}{\sin A \cos 2A + \sin 3A \cos 6A + \sin 4A \cos 13A} = \tan 9A.$$

Q15 The value of the expression

$$2 \cos\left(\frac{\pi}{13}\right) \cos\left(\frac{9\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right)$$

- (A) 0 (B) -1
(C) 1 (D) 2

Q16 If $4 \tan \theta \tan \phi = 3$, then $\frac{\cos(\theta - \phi)}{\cos(\theta + \phi)}$ is equal to

- (A) 1 (B) 3
(C) 4 (D) 7

Q17

$$\text{If } 3 \sin \alpha = 5 \sin \beta, \text{ then } \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}} =$$

- (A) 1 (B) 2
(C) 3 (D) 4

Q18

$$\frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ} =$$

- (A) $\tan 54^\circ$
(B) $\tan 36^\circ$
(C) $\tan 18^\circ$
(D) None of these

Q19

$$\frac{\sin(B + A) + \cos(B - A)}{\sin(B - A) + \cos(B + A)} =$$

- (A) $\frac{\cos B + \sin B}{\cos B - \sin B}$



- (B) $\frac{\cos A + \sin A}{\cos A - \sin A}$
(C) $\frac{\cos A - \sin A}{\cos A + \sin A}$
(D) None of these

Q20 If $\sin 2x = n \sin 2y$, then the value of $\frac{\tan(x+y)}{\tan(x-y)}$

is

- (A) $\frac{n+1}{n-1}$
(B) $\frac{n-1}{n+1}$
(C) $\frac{1-n}{n+1}$
(D) $\frac{1+n}{1-n}$



Answer Key

Q1 Proof
Q2 Proof
Q3 proof
Q4 proof
Q5 proof
Q6 proof
Q7 Proof
Q8 Proof
Q9 proof
Q10 proof

Q11 proof
Q12 proof
Q13 proof
Q14 proof
Q15 (A)
Q16 (D)
Q17 (D)
Q18 (A)
Q19 (B)
Q20 (A)



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